

48730 Cyber Security — Lab 1(Week 1&2)

48730 / 32548 Cyber Security — Lab 1 (Week 1 & 2)

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Task 1 — Sniff Login Details from Unencrypted HTTP Traffic

Wireshark captured (wifi promiscuous mode) traffic when login credentials were sent to <http://bias.csr. uni.bo.it/fvc2004/logon.asp> over http. Capture filtered with display filter frame contains "POST".

Question: Provide a screenshot of the username and password credentials used in Wireshark.

Answer: Because the site uses unencrypted HTTP, the login credentials are transmitted in clear text inside the HTTP POST body (HTML Form URL Encoded). The captured form items id and pwd are fully readable, as shown below.

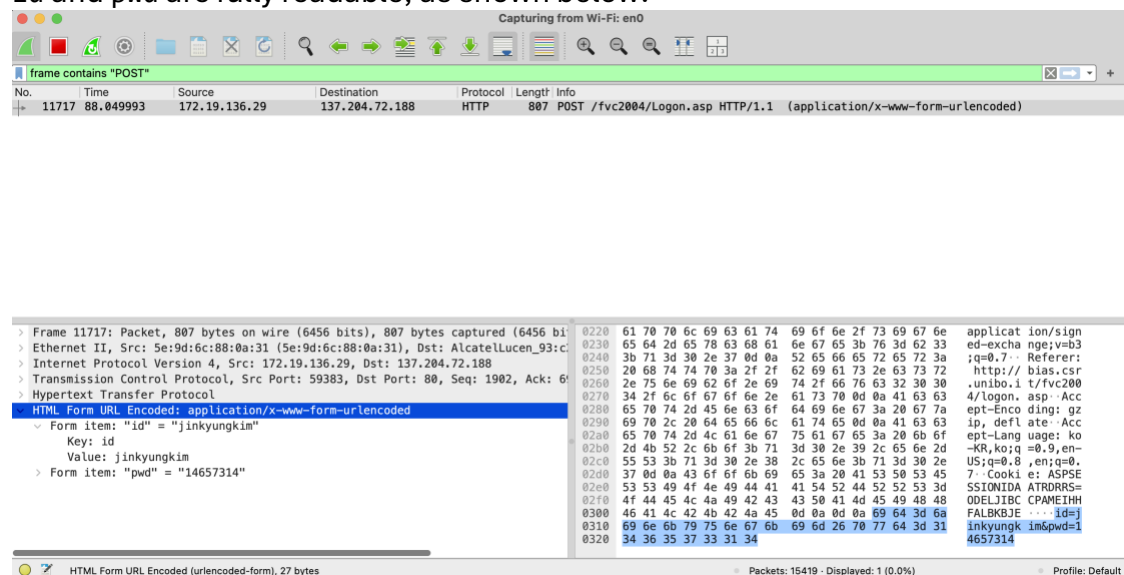


Fig 1. POST /fvc2004/Logon.asp — id = "jinkyungkim", pwd = "14657314" captured in clear text.

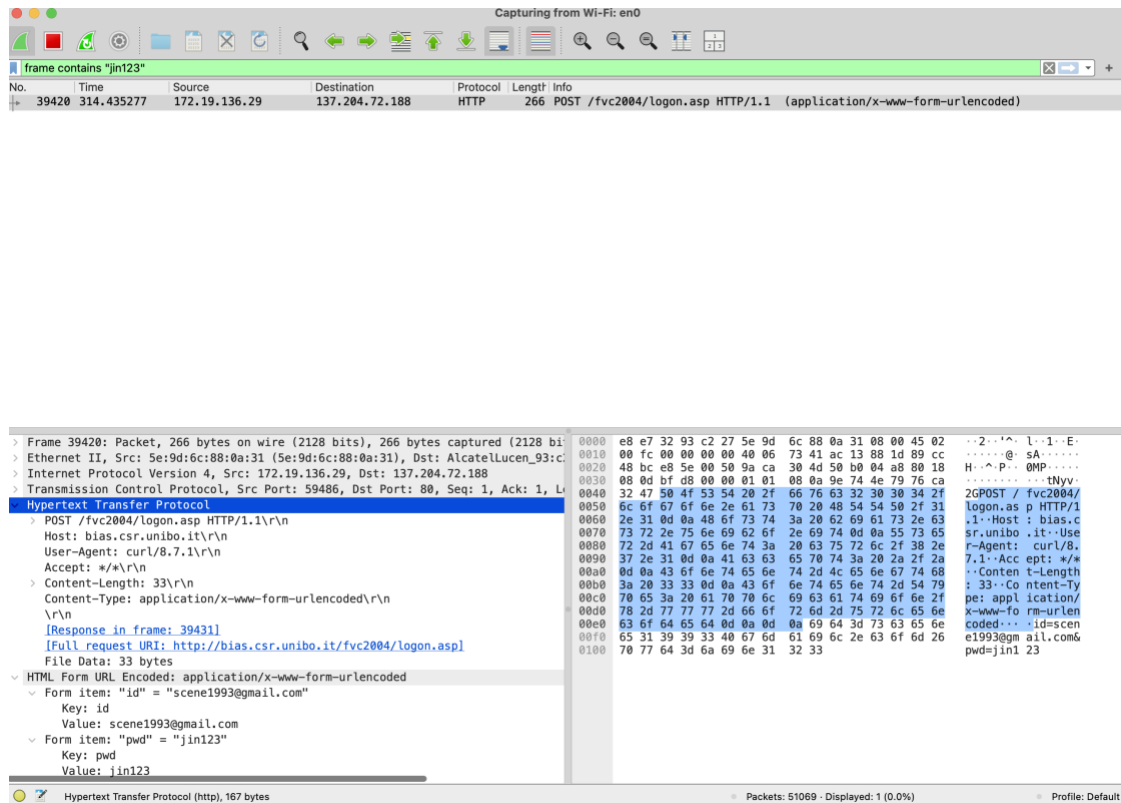


Fig 2. A second POST — id = "scene1993@gmail.com", pwd = "jin123" visible in plaintext.

Task 2 — Extract an Image from an Unencrypted HTTP Packet

Located GET packets with the same capture and exported objects through **File > Export Objects > HTTP**.

Question: Take a screenshot of the list of HTTP objects containing the list of images we got from the webpage.

Answer: The HTTP object list shows the image files served by bias.csr.unibo.it, including bullet.gif and fvc2004_head.gif (Content-Type image/gif), which can be

extracted with Save.

Wireshark · Export · HTTP object list

Text Filter: Content Type: All Content-Types

Packet	Hostname	Content Type	Size	Filename
1474	bias.csr.unibo.it	text/html	4943 bytes	logon.asp
1510	bias.csr.unibo.it	text/css	5630 bytes	fvc2004.css
1585	bias.csr.unibo.it	image/gif	820 bytes	bullet.gif
1588	biolab.csr.unibo.it	application/x-javascript	9919 bytes	cookies.js
1659	biolab.csr.unibo.it	text/css	2434 bytes	cookies.css
1662	bias.csr.unibo.it	text/html	1245 bytes	favicon.ico
11717	bias.csr.unibo.it	application/x-www-form-urlencoded	27 bytes	Logon.asp
11807	bias.csr.unibo.it	text/html	1208 bytes	Logon.asp
30470			1374 bytes	
30474			1374 bytes	
30475			1066 bytes	
39420	bias.csr.unibo.it	application/x-www-form-urlencoded	33 bytes	logon.asp
39431	bias.csr.unibo.it	text/html	1208 bytes	logon.asp
39464			1374 bytes	
39465			1374 bytes	
39484			1374 bytes	
39485			1374 bytes	
39486			1374 bytes	
39488			1374 bytes	
39490			1374 bytes	
39492			1374 bytes	
39493			1374 bytes	
39500			1374 bytes	
39503			232 bytes	
39533	bias.csr.unibo.it	image/gif	820 bytes	bullet.gif
39581	bias.csr.unibo.it	text/html	1245 bytes	fvc2004_head.gif

Help Preview Save All Close Save

Fig 3. HTTP object list — bullet.gif selected (image/gif, 820 bytes).

Wireshark · Export · HTTP object list

Text Filter: Content Type: All Content-Types

Packet	Hostname	Content Type	Size	Filename
1474	bias.csr.unibo.it	text/html	4943 bytes	logon.asp
1510	bias.csr.unibo.it	text/css	5630 bytes	fvc2004.css
1585	bias.csr.unibo.it	image/gif	820 bytes	bullet.gif
1588	biolab.csr.unibo.it	application/x-javascript	9919 bytes	cookies.js
1659	biolab.csr.unibo.it	text/css	2434 bytes	cookies.css
1662	bias.csr.unibo.it	text/html	1245 bytes	favicon.ico
11717	bias.csr.unibo.it	application/x-www-form-urlencoded	27 bytes	Logon.asp
11807	bias.csr.unibo.it	text/html	1208 bytes	Logon.asp
30470			1374 bytes	
30474			1374 bytes	
30475			1066 bytes	
39420	bias.csr.unibo.it	application/x-www-form-urlencoded	33 bytes	logon.asp
39431	bias.csr.unibo.it	text/html	1208 bytes	logon.asp
39464			1374 bytes	
39465			1374 bytes	
39484			1374 bytes	
39485			1374 bytes	
39486			1374 bytes	
39488			1374 bytes	
39490			1374 bytes	
39492			1374 bytes	
39493			1374 bytes	
39500			1374 bytes	
39503			232 bytes	
39533	bias.csr.unibo.it	image/gif	820 bytes	bullet.gif
39581	bias.csr.unibo.it	text/html	1245 bytes	fvc2004_head.gif

Help Preview Save All Close Save

Fig 4. HTTP object list — fvc2004_head.gif selected.

Task 3 — Information Gathering (target: www.uts.edu.au)

Netcraft Site Report (sitereport.netcraft.com) was used to gather information about www.uts.edu.au.

Network

Site	http://www.uts.edu.au	Domain	uts.edu.au
Netblock Owner	Cloudflare, Inc.	Nameserver	ns.uts.edu.au
Hosting company	Cloudflare	Domain registrar	audns.net.au
Hosting country	US	Nameserver organisation	whois.audns.net.au
IPv4 address	172.64.144.213 (VirusTotal)	Organisation	Unknown
IPv4 autonomous systems	AS13335	DNS admin	dnsadmin@uts.edu.au
IPv6 address	2606:4700:440b:0:0:0:ac40:90d5	Top Level Domain	Australia (.edu.au)
IPv6 autonomous systems	AS13335	DNS Security Extensions	Enabled
Reverse DNS	Unknown		

IP delegation

IPv4 address (172.64.144.213)

IP range	Country	Name	Description
::ffff:0:0:0:0/96	United States	IANA-IPv4-MAPPED-ADDRESS	Internet Assigned Numbers Authority
↳ 172.0.0.0–172.255.255.255	United States	NET172	Various Registries (Maintained by ARIN)
↳ 172.64.0.0–172.71.255.255	United States	CLOUDFLARENET	Cloudflare, Inc.
↳ 172.64.144.213	United States	CLOUDFLARENET	Cloudflare, Inc.

IPv6 address (2606:4700:440b:0:0:0:ac40:90d5)

IP range	Country	Name	Description
::/0	N/A	ROOT	Root inet6num object
↳ 2600::/12	United States	NET6-2600	American Registry for Internet Numbers
↳ 2606:4700::/32	United States	CLOUDFLARENET	Cloudflare, Inc.
↳ 2606:4700:440b:0:0:0:ac40:90d5	United States	CLOUDFLARENET	Cloudflare, Inc.

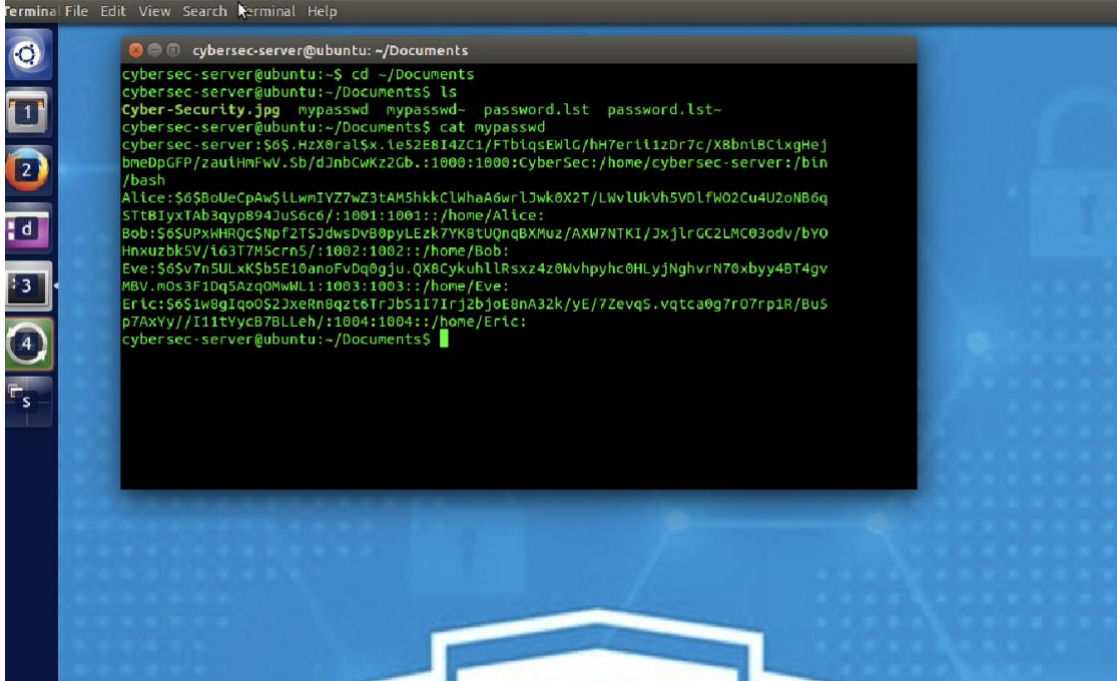
Fig 5. Netcraft site report for www.uts.edu.au.

1. **IP address:** 172.64.144.213 (IPv6: 2606:4700:440b::ac40:90d5)
2. **Type the IP in the browser — observations:** The UTS site does not load directly; a Cloudflare error / default page appears, because 172.64.144.213 is a Cloudflare proxy IP that routes by Host header/SNI, so the real origin is hidden.
3. **IP owner (Netblock owner):** Cloudflare, Inc.
4. **Server operating system:** Not disclosed — hidden behind the Cloudflare proxy.
5. **Web server type:** Cloudflare (nginx front-end).
6. **Server-side scripting technology:** Not disclosed (masked by Cloudflare).
7. **Domain admin email (possible phishing):** dnsadmin@uts.edu.au
8. **Reverse DNS:** Unknown.
9. **Domain registrar:** audns.net.au
10. **Nameserver organisation:** whois.audns.net.au (Nameserver: ns.uts.edu.au)
11. **Hosting company:** Cloudflare
12. **Hosting company location:** United States (US)

Task 4 — Password Cracking using John the Ripper

On the Cybersec-Server, the mypasswd file (username + hashed password) was cracked using single, wordlist and rules modes.

```
cd ~/Documents john --show mypasswd # 0 cracked (before) john mypasswd # single  
mode john --wordlist=password.lst mypasswd # wordlist mode john --  
wordlist=password.lst --rules mypasswd # mangling rules john --show mypasswd # final
```



The screenshot shows a terminal window titled "CyberSec-Server" with a menu bar (Terminal, File, Edit, View, Search, Terminal, Help). The terminal output is as follows:

```
cybersec-server@ubuntu: ~/Documents  
cybersec-server@ubuntu:~$ cd ~/Documents  
cybersec-server@ubuntu:~/Documents$ ls  
Cyber-Security.jpg mypasswd mypasswd- password.lst password.lst-  
cybersec-server@ubuntu:~/Documents$ cat mypasswd  
cybersec-server:$6$.HzX0ral$X.1e52E8I4ZC1/FTblqSEWLG/hH7er11zDr7c/X8bn18C1xgHej  
bmeDpGFP/zaulHmFwV.Sb/dJnbCwKz2Gb.:1000:1000:CyberSec:/home/cybersec-server:/bin  
/bash  
Alice:$6$BoUeCpAw$LLwnIYZ7wZ3tAM5hkkCLWhaA6wrLJwk0X2T/LWvLUkVh5VDLfw02Cu4U2oNB6q  
STtBIyxTAb3qyp894JuS6c6/:1001:1001:/home/Alice:  
Bob:$6$UPxHHRQc$Npf2TSJdwsDvB0pyLEzk7YK8tUQnqBXMuz/AXW7NTKI/JxJlrgCC2LMC03odv/bv0  
Hnxuzbk5V/i63T7M5crn5/:1002:1002:/home/Bob:  
Eve:$6$V7n5ULxK$b5E10anoFvDq0gju.QX8CykuhlLRsxz4z0Wvhyhc0HLyJNghvrN70xbyy48T4gv  
MBV.m0s3F1Dq5AzoQMwML1:1003:1003:/home/Eve:  
Eric:$6$1w0gIq052JxeRn8qzt6TrJb51I7Irj2bj0E8nA32k/yE/7ZevqS.vqtca0g7r07rp1R/Bu5  
p7AxYy//I11tYycB7BLLeh/:1004:1004:/home/Eric:  
cybersec-server@ubuntu:~/Documents$
```

Fig 6. Contents of mypasswd (hashed passwords).

```
cybersec-server@ubuntu: ~/Documents
cybersec-server@ubuntu:~$ cd ~/Documents
cybersec-server@ubuntu:~/Documents$ ls
Cyber-Security.jpg  mypasswd  mypasswd~  password.lst  password.lst~
cybersec-server@ubuntu:~/Documents$ cat mypasswd
cybersec-server:$6$.HzX0ral$X.1e52E8I4ZC1/FTbiqsEWlG/hH7eri1zDr7c/XBbniBCixgHej
bmeDpGFP/zauHMFwV.5b/dJnbCwKz2Gb.:1000:1000:cybersec:/home/cybersec-server:/bin
/bash
Alice:$6$BoUeCpAw$ilWmIYZ7wZ3tAM5hkkClWhaA6wrlJwk0X2T/LWv1ukVhSVDlfW02Cu4U2oNB6q
STtBIyxTAB3qyp894Ju56c6/:1001:1001:/home/Alice:
Bob:$6$UPxWHRQc$Npf2TSJdwsDvB0pyLEzk7YK8tUQnqBXMuz/AXW7NTKI/JxjlrGC2LMC03odv/bYO
Hnxuzbk5V/i63T7M5crn5/:1002:1002:/home/Bob:
Eve:$6$V7n5ULxK$b5E10anoFvDq0gju.QX8CykuhlLRsxz4z0Wvhpyhc0HlyjNghvrN70xbyy4BT4gv
MBV.m0s3F1Dq5Azq0MwHL1:1003:1003:/home/Eve:
Eric:$6$1w8gIqo0$2JxeRn8qzt6TrJbs1I7Irj2bjoE8nA32k/yE/7ZevqS.vqtca0g7r07rp1R/Bu5
p7AxYy//I11tYycB7BLLeH/:1004:1004:/home/Eric:
cybersec-server@ubuntu:~/Documents$ john --show mypasswd
0 password hashes cracked, 5 left
cybersec-server@ubuntu:~/Documents$
```

Fig 7. john --show before cracking: 0 cracked, 5 left.

```
cybersec-server@ubuntu: ~/Documents
Bob:$6$UPxWHRQc$Npf2TSJdwsDvB0pyLEzk7YK8tUQnqBXMuz/AXW7NTKI/JxjlrGC2LMC03odv/bYO
Hnxuzbk5V/i63T7M5crn5/:1002:1002:/home/Bob:
Eve:$6$V7n5ULxK$b5E10anoFvDq0gju.QX8CykuhlLRsxz4z0Wvhpyhc0HlyjNghvrN70xbyy4BT4gv
MBV.m0s3F1Dq5Azq0MwHL1:1003:1003:/home/Eve:
Eric:$6$1w8gIqo0$2JxeRn8qzt6TrJbs1I7Irj2bjoE8nA32k/yE/7ZevqS.vqtca0g7r07rp1R/Bu5
p7AxYy//I11tYycB7BLLeH/:1004:1004:/home/Eric:
cybersec-server@ubuntu:~/Documents$ john --show mypasswd
0 password hashes cracked, 5 left
cybersec-server@ubuntu:~/Documents$ john mypasswd
Loaded 5 password hashes with 5 different salts (crypt, generic crypt(3) [?/64])
Press 'q' or Ctrl-C to abort, almost any other key for status
cybersec      (cybersec-server)
12345         (Bob)
password      (Alice)
3g 0:00:02:46 8% 2/3 0.01797g/s 93.96p/s 168.1c/s 168.1C/s newpass1..pepper1
Use the "--show" option to display all of the cracked passwords reliably
Session aborted
cybersec-server@ubuntu:~/Documents$ john --show mypasswd
cybersec-server:cybersec:1000:1000:cybersec:/home/cybersec-server:/bin/bash
Alice:password:1001:1001:/home/Alice:
Bob:12345:1002:1002:/home/Bob:

3 password hashes cracked, 2 left
cybersec-server@ubuntu:~/Documents$
```

Fig 8. Single mode cracks cybersec-server, Bob, Alice → 3 cracked.


```

cybersec-server@ubuntu:~$ cd ~/Documents
cybersec-server@ubuntu:~/Documents$ ls
Cyber-Security.jpg mypasswd mypasswd- password.lst password.lst-
cybersec-server@ubuntu:~/Documents$ cat mypasswd
cybersec-server:$6$Hx0ral$X.1e52EB14ZC1/FTblqEHLG/hH7erit1zDr7c/XBbntBCLxgHej
bneDpGFP/zaulHmFwW.5b/dJnbCwKz2Gb.:1000:1000:cyberSec:/home/cybersec-server:/bin
/bash
Alice:$6$5B0JecPaw$1LwmIVZ7wZ3tAM5hkkCLWhaA6wrLJwk0X2T/LHvLukVh5VDLfW02Cu4U2oNB6q
STtBIyxTab3qyp894Ju56c6/:1001:1001:./home/Alice:
Bob:$6$SUPuWHRQc$Npf2T5JdwsDvB0pyLEzk7YK8tUQnqBXMuz/AXu7NTKI/3xjLrGC2LMC03odv/bYO
HnxuzbkSV/i63T7MScrn5/:1002:1002:./home/Bob:
Eve:$6$5V7n5ULXK$5E10anoFvDq0gju.QX8CykuhlLRsxz4z0Wvhpyhc0HlyjHghvrN70xbyy4BT4gv
MBV.m0s3F1Dq5AzaQ0HwL1:1003:1003:./home/Eve:
Eric:$6$1w8Iqo052JxeRn8qzt6TrJb5117I.rJ2bjoE8nA32k/yE/7Zevq5.vqtca0g7r07rp1R/Bu5
o7AXYy//I11tYyc87BLLeh/:1004:1004:./home/Eric:
cybersec-server@ubuntu:~/Documents$ john --show mypasswd
0 password hashes cracked, 5 left
cybersec-server@ubuntu:~/Documents$ john mypasswd
Loaded 5 password hashes with 5 different salts (crypt, generic crypt(3) [?/64])
Press 'q' or Ctrl-C to abort, almost any other key for status
cybersec (cybersec-server)
12345 (Bob)
password (Alice)
3g 0:00:02:46 8% 2/3 0.01797g/s 93.96p/s 168.1c/s 168.1c/s newpass1.pepper1
Use the "--show" option to display all of the cracked passwords reliably
Session aborted
cybersec-server@ubuntu:~/Documents$ john --show mypasswd
cybersec-server:cybersec:1000:1000:cyberSec:/home/cybersec-server:/bin/bash
Alice:password:1001:1001:./home/Alice:
Bob:12345:1002:1002:./home/Bob:

3 password hashes cracked, 2 left
cybersec-server@ubuntu:~/Documents$ john --wordlist=password.lst --rules mypasswd
Loaded 5 password hashes with 5 different salts (crypt, generic crypt(3) [?/64])
Remaining 2 password hashes with 2 different salts
Press 'q' or Ctrl-C to abort, almost any other key for status
Pa$$w0rd (Eve)
1g 0:00:01:21 9% 0.01231g/s 166.7p/s 167.9c/s 167.9c/s Candy1..Marlboro1
Use the "--show" option to display all of the cracked passwords reliably
Session aborted
cybersec-server@ubuntu:~/Documents$ john --wordlist=password.lst mypasswd
Loaded 5 password hashes with 5 different salts (crypt, generic crypt(3) [?/64])
Remaining 1 password hash
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:23 100% 0g/s 150.5p/s 150.5c/s 150.5c/s 00812..sss
Session completed
cybersec-server@ubuntu:~/Documents$ ^C
cybersec-server@ubuntu:~/Documents$ john --show mypasswd
cybersec-server:cybersec:1000:1000:cyberSec:/home/cybersec-server:/bin/bash
Alice:password:1001:1001:./home/Alice:
Bob:12345:1002:1002:./home/Bob:
Eve:Pa$$w0rd:1003:1003:./home/Eve:

4 password hashes cracked, 1 left
cybersec-server@ubuntu:~/Documents$

```

Fig 9. After wordlist + rules: cybersec, Alice=password, Bob=12345, Eve=Pa\$\$w0rd (4 cracked, 1 left).

Cracked passwords: cybersec-server = cybersec, Alice = password, Bob = 12345, Eve = Pa\$\$w0rd.

Challenge 1: Cracking Eric's password

Answer: Unable to crack Eric's password using single, wordlist or rules modes in a reasonable time frame. Eric used a good password that wasn't found in the dictionary and that can't be cracked with mangling rules (only an exhaustive brute-force / incremental attack would discover it).

Challenge 2: How to secure the password

Answer: Weak, short or dictionary-based passwords (e.g. password, 12345) are cracked almost instantly, while a password that is not in any wordlist (Eric's) survives. A secure password should be long (12+ characters), mix upper/lower case, numbers and symbols, avoid dictionary words and personal information, and be unique per account.

Task 5 — SQL Injection (login bypass at http://localhost/)

The login form builds the query `SELECT username,password FROM members WHERE username='$u' AND password='$p'`, with no input sanitisation, so it is vulnerable to SQL injection.

I. Username = 123456789, Password = SQL command

Username: 123456789 Password: ' OR '1'='1

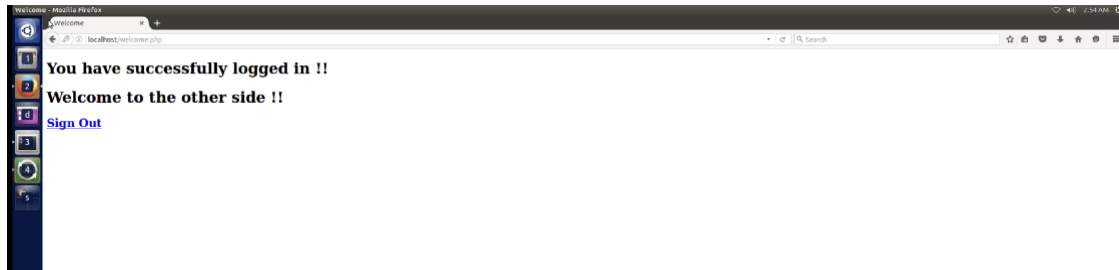


Fig 10. Login bypassed — "You have successfully logged in".

II. Both Username and Password as SQL commands

Username: ' OR '1'='1 Password: ' OR '1'='1

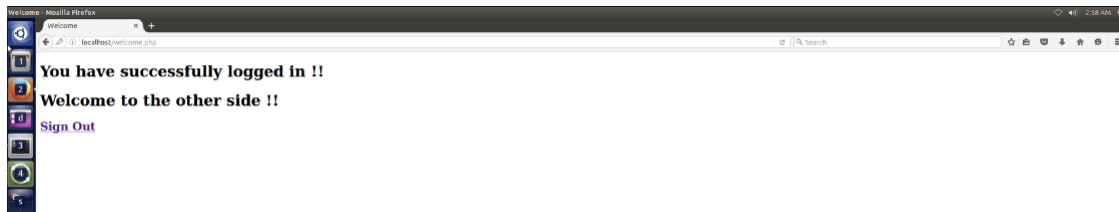


Fig 11. Login bypassed with injection in both fields.

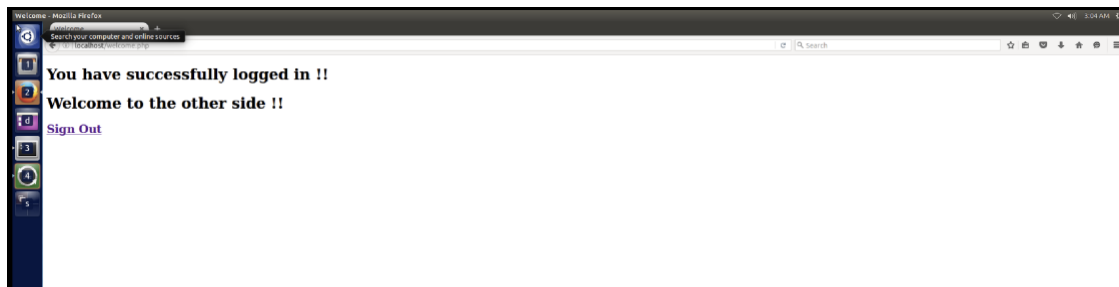


Fig 12. Additional successful bypass.

Q1: Provide screenshots and mention the SQL commands used to gain unauthorized access.

Answer: The payload ' OR '1'='1 makes the WHERE clause always true, so the query returns rows and authentication is bypassed (see Figs 10–12). The commands used are listed under each sub-task above.

Note: Sub-tasks III (dumping all usernames/passwords) and IV (logging in as a specific extracted user) rely on the application echoing query data. The welcome page only prints a static "successfully logged in" message and does not display returned rows, so login-bypass injection was demonstrated successfully (I & II); full data extraction was not displayed by the application.

Q2: Based on your observations, suggest a defence.

Answer: Use parameterized queries / prepared statements so user input is never concatenated into SQL; validate and sanitise all input; apply least-privilege database accounts; disable detailed SQL error messages to the client; and use a web application firewall (WAF).

Task 6 — Port Scanning (aggressive scan from Cybersec-Attacker)

Q1: Run an aggressive port scan on cybersec-server (10.0.2.6) and cybersec-client (10.0.2.8) from cybersec-attacker. Provide screenshots of the port scan report.

```
sudo nmap -A 10.0.2.6 # cybersec-server sudo nmap -A 10.0.2.8 # cybersec-client
```

Answer: Both hosts show the same open services: 21/tcp ftp (vsftpd 3.0.2), 22/tcp ssh (OpenSSH), 23/tcp telnet, 53/tcp domain (BIND 9.9.5), 80/tcp http (Apache

```

2 cybersec-attacker@ubuntu:~$ sudo nmap -A 10.0.2.6
[sudo] password for cybersec-attacker:

Starting Nmap 6.40 ( http://nmap.org ) at 2026-08-12 04:21 PDT
Nmap scan report for 10.0.2.6
Host is up (0.00072s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.2
22/tcp    open  ssh      (protocol 2.0)
|_ ssh-hostkey: 1024 d8:6c:f3:db:fd:3a:79:93:b0:41:45:e2:b5:80:c9:0b (DSA)
|_ 2048 ce:87:27:73:d0:bd:bf:4b:7a:92:96:3b:bc:1d:18:8b (RSA)
|_ 256 9a:97:a9:8f:03:2e:aa:66:fc:d0:6e:26:ed:e9:10:7b (ECDSA)
23/tcp    open  telnet   Linux telnetd
53/tcp    open  domain
|_ dns-nsid:
|_ bind.version: 9.9.5-3ubuntu0.8-Ubuntu
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
|_ http-methods: No Allow or Public header in OPTIONS response (status code 200)
|_ http-title: Login Page
1 service unrecognized despite returning data. If you know the service/version,
please submit the following fingerprint at http://www.insecure.org/cgi-bin/servicefp-submit.cgi :
SF-Port22-TCP:V=6.40%I=7%D=8/12%T=6A7C573C%P=x86_64-pc-linux-gnu%r(NULL
SF:,2B:,"SSH-2.0-OpenSSH_6\..6\..1pI\X20Ubuntu-2ubuntu2\..7\r\n");
MAC Address: 66:60:18:56:2B:9C (Unknown)
No exact OS matches for host (If you know what OS is running on it, see http://nmap.org/submit/).
TCP/IP fingerprint:
OS:SCAN(V=6.40%E=4%D=8/12%OT=21%CT=1%CU=37503%PV=Y%D=1%DC=D%G=Y%M=666018%T
OS:M=6A7C5754%P=x86_64-pc-linux-gnu)SEQ(SP=102%GCD=1%ISR=10B%TI=Z%CI=1%TS=8
OS:SEQ(SP=102%GCD=1%ISR=10B%TI=Z%CI=1%II=1%TS=8)OPS(O1=M5B4S11NW7%O2=M5B4
OS:ST11NW7%O3=M5B4NN11NW7%O4=M5B4ST11NW7%O5=M5B4ST11NW7%O6=M5B4ST11)WIN(W1
OS:=7120%W2=7120%W3=7120%W4=7120%W5=7120%W6=7120)ECN(R=Y%DF=Y%T=40%W=7210%O
OS:=M5B4NN5NW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%W=0%S=A%S+F=A%RD=0%Q=)T2(R=N)T3(R=N
OS:)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=
OS:S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF
OS:=Y%T=40%W=0%S=Z%A=S+F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=
OS:G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 0.72 ms 10.0.2.6

OS and Service detection performed. Please report any incorrect results at http://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 34.05 seconds

```

```
Nmap done: 1 IP address (1 host up) scanned in 34.63 seconds
cybersec-attacker@ubuntu:~$ sudo nmap -A 10.0.2.8

Starting Nmap 6.40 ( http://nmap.org ) at 2026-08-12 04:22 PDT
Nmap scan report for 10.0.2.8
Host is up (0.0025s latency).
Not shown: 995 closed ports
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.2
22/tcp    open  ssh      (protocol 2.0)
|_ ssh-hostkey: 1024 d8:6c:f3:db:fd:3a:79:93:b0:41:45:e2:b5:80:c9:0b (DSA)
|_ 2048 ce:87:27:73:d0:bd:bf:4b:7a:92:96:3b:bc:1d:18:8b (RSA)
|_ 256 9a:97:a9:8f:03:2e:aa:66:fc:d0:6e:26:red:e9:10:7b (ECDSA)
23/tcp    open  telnet   Linux telnetd
53/tcp    open  domain
|_ dns-nsid:
|_ bind.version: 9.9.5-3ubuntu0.8-Ubuntu
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
|_ http-methods: No Allow or Public header in OPTIONS response (status code 200)
|_ http-title: Login Page
1 service unrecognized despite returning data. If you know the service/version,
please submit the following fingerprint at http://www.insecure.org/cgi-bin/servicefp-submit.cgi :
SF-Port22-TCP:V=6.40%I=7%D=8/12%Time=6A7C5770%P=x86_64-pc-linux-gnu%r(NULL
SF:,2B,"SSH-2\0-OpenSSH_6\0\0.1p1\0x20Ubuntu-2ubuntu2\0.7\r\n");
MAC Address: 36:7E:09:64:6D:F3 (Unknown)
No exact OS matches for host (If you know what OS is running on it, see http://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=6.40%E=4%D=8/12%OT=21%CT=1%CU=38745%PV=Y%D5=1%DC=D%G=Y%M=367E09%T
OS:M=6A7C5788%P=x86_64-pc-linux-gnu)SEQ(SP=107%GCD=1%ISR=10D%TI=Z%CI=I%II=I
OS:%TS=8)SEQ(SP=107%GCD=1%ISR=10D%TI=Z%CI=RD%TS=8)OPS(O1=M5B4ST11NW7%O2=M5B
OS:4ST11NW7%O3=M5B4NNT11NW7%O4=M5B4ST11NW7%O5=M5B4ST11NW7%O6=M5B4ST11)WIN(W
OS:1=7120%W2=7120%W3=7120%W4=7120%W5=7120%W6=7120)ECN(R=Y%DF=Y%T=40%W=7210%
OS:O=M5B4NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=
OS:N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A
OS:=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%D
OS:F=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL
OS:=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT ADDRESS
1 2.52 ms 10.0.2.8

OS and Service detection performed. Please report any incorrect results at http://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 34.28 seconds
cybersec-attacker@ubuntu:~$
```

Fig 14. Aggressive scan of cybersec-client (10.0.2.8).